BUDDAREDDY DIVIJ

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Professional Summary

B.Tech graduate in Artificial Intelligence and Machine Learning with hands-on experience developing software applications and machine learning solutions using Python. Experienced in application development, data Preprocessing, testing, debugging, model development, and deployment through internship and academic projects.

Education

Kalasalingam Academy of Research and Education, Tamil Nadu 2022 – 2026
Bachelor of Technology in Artificial Intelligence & Machine Learning CGPA: 7.8

Experience

Software and Application Developer *AI Automation Labs*
May 2026 – Present

- Developed software and machine learning applications using Python, including data preprocessing, feature engineering, model development, testing, validation, and deployment.
- Contributed to end-to-end application development by implementing, testing, debugging, and improving Python-based solutions for business-focused use cases.

Technical Skills

Programming Languages: Python (OOP), SQL
Frameworks & Databases: FastAPI, PostgreSQL, REST APIs
AI & Machine Learning: Large Language Models (LLM), Retrieval-Augmented Generation (RAG), LLM Fine-tuning, Reinforcement Learning, Supervised Learning, Unsupervised Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Machine Learning
Libraries: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
Development Practices: Data Analysis, Data Preprocessing, Feature Engineering, Model Evaluation, Prompt Engineering, Model Deployment, Testing, Debugging
Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Hugging Face

Projects

AI-Powered Robotic Arm *[Python, C++, Computer Vision]*

- Developed an AI-powered robotic arm application for real-time color-based object detection and automated pick-and-place operations.
- Implemented computer vision-based object detection and classification for red, green, and blue objects using live camera input.

NeuroMap AI System *[TypeScript, React, Python, Machine Learning, Three.js, Google Gemini]*

- Developed an end-to-end neuroimaging research platform for MRI/dMRI analysis, quantitative brain morphometry, structural connectome analysis, and machine learning-based research insights.
- Implemented multi-model ML benchmarking, SHAP-based feature attribution, 3D Grad-CAM visualization, interactive 3D brain connectivity maps, and Gemini-powered research report generation.

StartupOps AI *[Python, FastAPI, Next.js, LangGraph, PostgreSQL]*

- Developed an AI-powered operations platform for startups that coordinates specialized agents to investigate cross-system issues, prioritize actions, and generate evidence-backed recommendations.
- Implemented a multi-agent architecture with supervisor, planner, research, investigation, signal, action, and verification agents, integrating persistent workspace state, approval workflows, and secure external-system access.

MotherCare Platform *[JavaScript, HTML, CSS, RAG, Google Gemini, Semantic Search]*

- Developed a multilingual maternity and parenting AI platform using RAG and Google Gemini to provide evidence-grounded pregnancy, postpartum, and newborn-care information.
- Implemented emergency red-flag detection, stage-specific context selection, source provenance, and an admin dashboard for consultation analytics and content-gap identification.

Certifications

- Python – HackerRank
- AI Agents Fundamentals – Hugging Face